

COURSE MATERIAL

SUBJECT	ENGINEERING GRAPHICS (ME23AES102)
UNIT	4
COURSE	B.TECH
SEMESTER	11
DEPARTMENT	MECHANICAL ENGINEERING
PREPARED BY (Faculty Name/s)	Dr. N RAJESH, Associate Professor
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1. COURSE OBJECTIVES

The objectives of this course is to introduce and make the students

- To enable with various concepts like dimensioning, conventions and standards related to Engineering Drawing
- To impart knowledge on the Projection of Points, Lines and Plane surfaces
- To improve the visualization skills for better understanding of projection of solids
- To develop the imaginative skills required to understand Section of solids and Developments of surfaces.
- To understand the viewing perception of a solid object in Isometric and orthographic projections.

2. PREREQUISITES

Students should have knowledge on
Basic Mathematics

3. SYLLABUS

UNIT IV

Sections of Solids: Perpendicular and Inclined Section Planes, Sectional Views and True Shape of Section, Sections of Solids in Simple Position only.

Development of Surfaces: Methods of Development: Parallel Line Development and Radial Line Development. Development of a Cube, Prism, Cylinder, Pyramid and Cone.

4. COURSE OUTCOMES

After completing the course, the student will be able to

1. Draw various engineering curves, scales. (L3)
2. Draw and Interpret orthographic projections of points, lines, planes. (L3)
3. Draw the projection of solids in various positions. (L3)
4. Draw and Explore the sections of solids and development of surfaces. (L3)
5. Draw an isometric and orthographic views of simple solids. (L3).

5. Co-PO / PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	P10	PO11	PO12	PSO1	PSO2
CO1	2	2								2		2	2	2
CO2	2	2								2		2	2	2
CO3	2	2								2		2	2	2
CO4	2	2								2		2	2	2
CO5	2	2			1					2		2	2	2

6. LESSON PLAN

LECTURE	WEEK	TOPICS TO BE COVERED	REFERENCES
1	1	Introduction to the projection of solids – I	T1
2		Lab practice	T1, R1
3		Introduction to the projection of solids – II	T1, R1
4		Lab practice	T1, R2
5	2	Development of Surfaces	T1, R1
6		Lab practice	T2, R1



7. LECTURE NOTES

4.1 SECTIONS OF SOLIDS

4.1.1 Sections of Solids:

Sections and sectional views are used to show hidden detail more clearly. They are created by using a cutting plane to cut the object.

A section is a view of no thickness and shows the outline of the object at the cutting plane. Visible outlines beyond the cutting plane are not drawn.

A sectional view, displays the outline of the cutting plane and all visible outlines which can be seen beyond the cutting plane.

Improve visualization of interior features. Section views are used when important hidden details are in the interior of an object. These details appear as hidden lines in one of the orthographic principal views; therefore, their shapes are not very well described by pure orthographic projection.

4.1.2 Cutting Plane:

Section views show how an object would look if a cutting plane (or saw) cut through the object and the material in front of the cutting plane was discarded.

5.3 Representation of cutting plane:

According to drawing standards cutting plane is represented by chain line with alternate long dash and dot. The two ends of the line should be thick.

I. Hatching:

On sections and sectional views solid area should be hatched to indicate this fact. Hatching is drawn with a thin continuous line, equally spaced (preferably about 4mm apart, though never less than 1mm) and preferably at an angle of 45 degrees.

Problem 1: A square prism of base side 30 mm and axis length 60 mm is resting on HP on one of its bases, with a base side inclined at 30° to VP. It is cut by a plane inclined at 10° to HP and perpendicular to VP and is bisecting the axis of the prism. Draw its front view, sectional top view and true shape of section.

Ans:

Draw the projections of the prism in the given position. The top view is drawn and the front view is projected.

To draw the cutting plane, front view and sectional top view

1. Draw the Vertical Trace (VT) of the cutting plane inclined at 40° to XY line and passing through the mid point of the axis.
2. As a result of cutting, longer edge $a'p'$ is cut, the end a' has been removed and the new corner $1'$ is obtained.
3. Similarly $2'$ is obtained on longer edge $b'q'$, $3'$ on $c'r'$ and $4'$ on $d's'$,
4. Show the remaining portion in front view by drawing dark lines.
5. Project the new points $1', 2', 3'$ and $4'$ to. get 1, 2, 3 and 4 in the top view of the prism, which are coinciding with the bottom end of the longer edges p, q, r and s respectively.
6. Show the sectional top view or apparent section by joining 1, 2, 3 and 4' by drawing hatching lines.

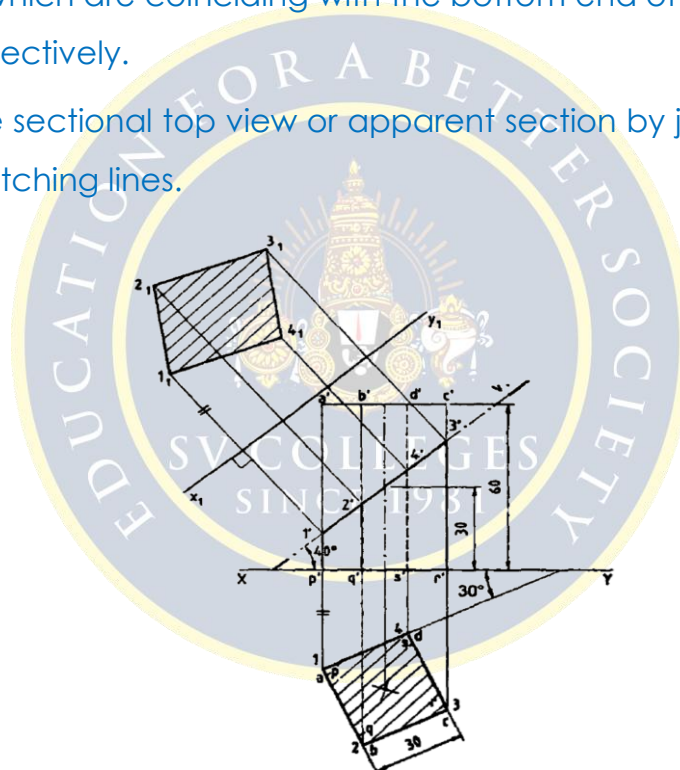


Figure 1.1

1. Consider an auxiliary inclined plane parallel to the cutting plane and draw the new reference line $x_1 y_1$ parallel to VT of the cutting plane at an arbitrary distance from it.
2. Draw projectors passing through $1', 2', 3'$ and $4'$ perpendicular to $x_1 y_1$ line.
3. The distance of point 1 in top view from XY line is measured and marked from $x_1 y_1$ in the projector passing through $1'$ to get $11'$. This is repeated to get the other points $21, 31$ and 41

4. Join these points to get the true shape of section as shown by drawing the hatching lines.

Problem : A pentagonal pyramid of base side 40 mm and axis length 80mm is resting on HP on its base with one of its base side parallel to VP. It is cut by a plane inclined at 30° to HP and perpendicular to VP and is bisecting the axis. Draw its front view, sectional top view, and the true shape of section.

Ans:

To draw the cutting plane, front view and sectional top view

1. Draw the VT of the cutting plane inclined at 30° to XY line and passing through the midpoint of J the axis.
2. As a result of cutting, new comers 1', 2', 3', 4' and 5' are obtained on slant edges a '0', b '0', c '0', d'o' and e '0' respectively.
3. Show the remaining portion in front view by drawing dark lines.
4. Project the new points to get 1,2,3,4 and 5 in the top view on the respective slant edges.
5. Note that 2' is extended horizontally to meet the extreme slant edge a '0 ' at m', it is projected to meet ao in top view at m. Considering 0 as centre, om as radius, draw an arc to get 2 on bo.

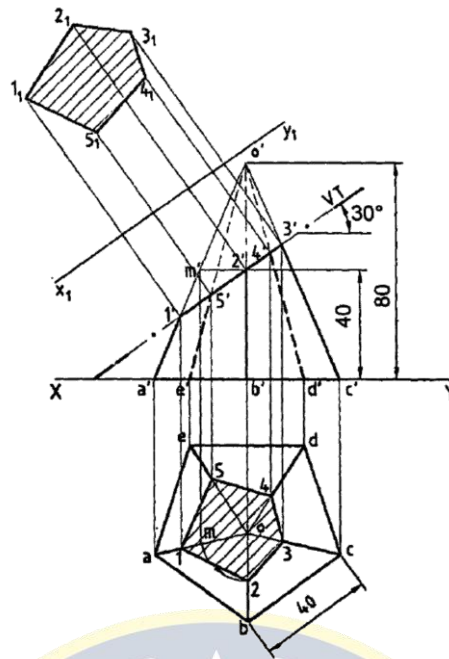


Figure 1.2

6. Join these points and show the sectional top view by drawing hatching lines. SS

To draw true shape of section.

1. Draw the new reference line X_1Y_1 parallel to VT of the cutting plane.
2. Projectors from $1', 2'$ etc. are drawn perpendicular to X_1Y_1 line.
3. The distance of point 1 in top view from XY line is measured and marked from X_1Y_1 in the projector passing through $1'$ to get $1''$. This is repeated to get $2'', 3''$ etc.
4. Join these points and draw hatching lines to show the true shape of section.

Problem: A cylinder of base diameter 40 mm and height 60 mm rests on its base on HP. It is cut by a plane perpendicular to VP and inclined at 30° to HP and meets the axis at a distance 30 mm from base. Draw the front view, sectional top view, and the true shape of section.

Ans:

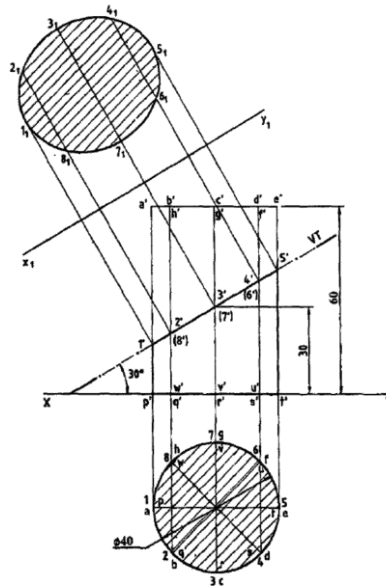


Figure 1.3

1. Draw the projections of the cylinder. The top view is drawn and the front view is projected. Consider generators by dividing the circle into equal number of parts and project them to the front view.
2. To draw the cutting plane, front view and sectional top view
3. 1. Draw the VT of the cutting plane inclined at 30° to XY line and passing through a point on the axis at a distance 30 mm from base.
4. 2. The new point 1', 2' etc. are marked on the generators a' p', h' q' etc.
5. 3. Project the new points to the top view to get 1, 2, etc. which are coinciding with p, q, etc. on
6. the base circle.

To draw true shape of section.

1. Draw X1 Y1 line parallel to VT of the cutting plane.
2. Draw the projectors through 1', 2', etc. perpendicular to X1 Y1 line.
3. The distance of point 1 in top view from XY line is measured and marked from X1 Y1 in the projector passing through 1' to get 11'. This is repeated to get other points 21, 31 etc.
4. Join these points by drawing smooth curve to get the true shape of section and this is shown by hatching lines.

Problem : A cone of base diameter 50 mm and axis length 75 mm, resting on HP on its base is cut by a plane inclined at 45° to HP and perpendicular to VP and is bisecting the axis. Draw the front view and sectional top view and true shape of this section.

Ans:

Draw the projections of the cone. Consider generators by dividing the circle into equal number of parts and project them to the front view.

To draw the cutting plane, front view, and sectional top view

1. Draw the VT of the cutting plane inclined at 45° to the XY line and passing through the midpoint of the axis.
2. New points 1', 2' etc; are marked on the generators a' 0', h' 0', etc.
3. Project the new points to the top view to get 1, 2, etc. on the generators ao, bo etc.
4. Note that the new point 3' is produced to mark m' on a' 0' and is projected to get m on ao. Considering O as centre and om as radius, draw an arc to get n on co in the top view. The same method is repeated to get 7 on go.
5. Join these points by drawing smooth curve and draw the hatching lines to show the sectional, top view.

To draw true shape of section

1. Draw X1 Y1 line parallel to VT of the cutting plane.
2. Draw the projectors through 1', 2' etc. perpendicular to X1 Y1 line.
3. The distance of point 1 in top view from XY line is measured and marked from X1 Y1 in the projector passing through l' to get 11 and is repeated to get 21 ' 31 etc.

Join these points by drawing smooth curve to get the true shape of section and is shown by hatching lines.

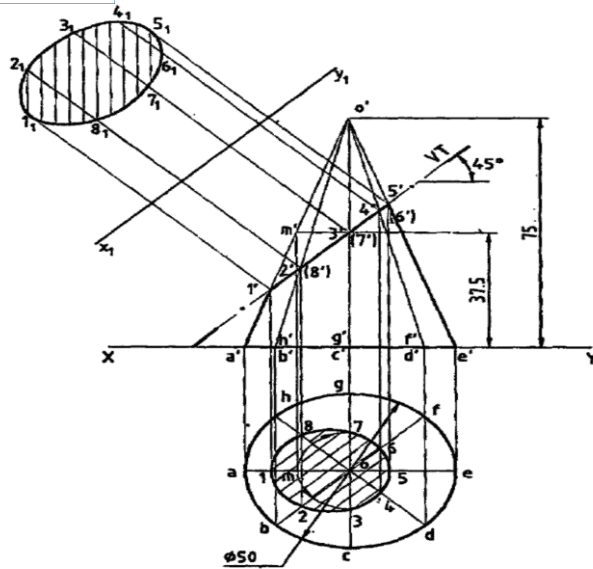


Figure 1.4



DEVELOPEMENT OF SURFACES

1. A cylinder of base diameter 40mm and axis 60mm long is resting on H.P. such that its axis is parallel to V.P. and perpendicular to H.P. Draw its Development. (The views are given in Third angle projection)

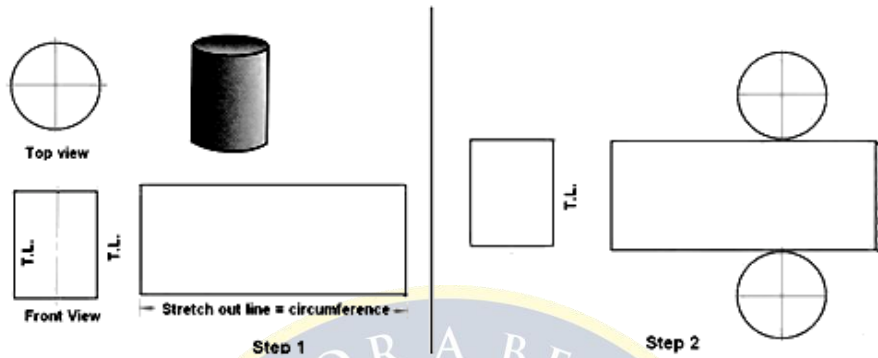


Figure 1.1

2. A square prism of side of base 40mm and axis 80mm long, is resting on its base on H.P. such that, a rectangular face of it is parallel to V.P. Draw the development of the prism.

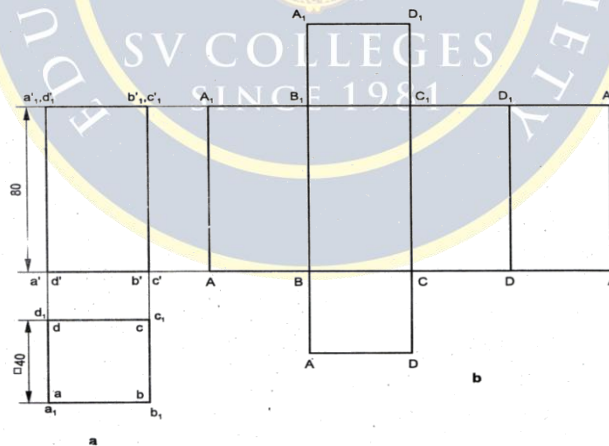


Figure 1.2

Construction:

1. Draw the Projections of the Prism.
2. Draw the stretch – out line AA and mark – off the sides of the base along this line in succession. i.e., AB, BC, CD and DA.
3. Erect perpendiculars through these points and mark the edges AA1, BB1 etc.,

4. Add the bases ABCD and A1 B1 C1 D1 suitably.

Note:

- Stretch-out line is drawn in-line with the base in the front view, to complete the development quickly.
 - All the lines on the development should represent the true lengths.
3. A pentagonal pyramid of side of base 30mm and 60mm long, is resting on its base on H.P. with an edge of the base parallel to V.P. Draw the development of the lateral surface of the pyramid.

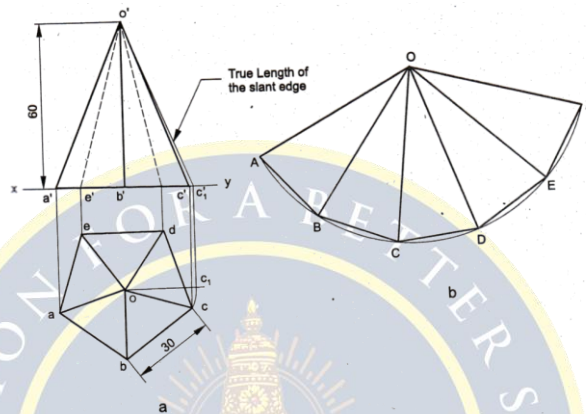


Figure 1.3

Construction:

1. Draw the projections of the pentagonal pyramid.
2. Determine the true length of the slant edge OC of the pyramid as shown.
3. With any point O as centre and radius equal to the true length of the slant edge, draw an arc of a circle.
4. With radius equal to the side of the base, step-off five divisions on the above arc.
5. Join the above division points in the order and also with the centre of the arc.
6. The figure thus formed is the development of the surface of the pyramid.

Hint:

- The development of a pyramid consists of a number of equal isosceles triangles. The base of the triangle is equal to the edge of the base and the sides equal to the slant height of the pyramid respectively.

- The true length of the slant edge may be measured from the front view, by making its top view parallel to xy .
4. Draw the development of surface of the square pyramid of base 30mm and axis 50mm when it is resting on its square base such that two of its side are perpendicular to V.P.

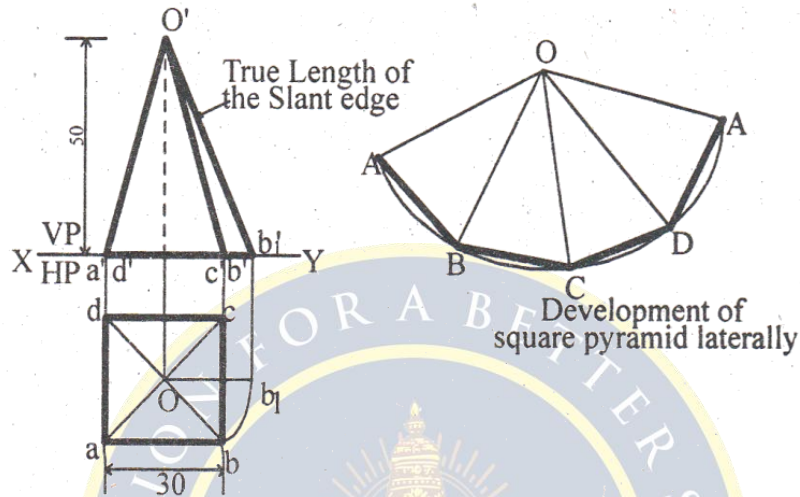


Figure 1.4

Construction:

1. Draw the top view of the square pyramid of base 30mm.
2. Project top view and draw front view of the pyramid of height 50mm.
3. Determine the true length of the slant edge of the pyramid.
4. With any point O as centre and radius equal to the true length of the slant edge, draw an arc of the circle.
5. With radius equal to the side of the base, step- off four divisions on the above arc.
6. Join the above division points in the order and also with the centre of the arc. The figure thus formed is the development of the lateral surface of the pyramid.
5. Draw the development of surface of a cone of base diameter 20mm and axis height 30mm.

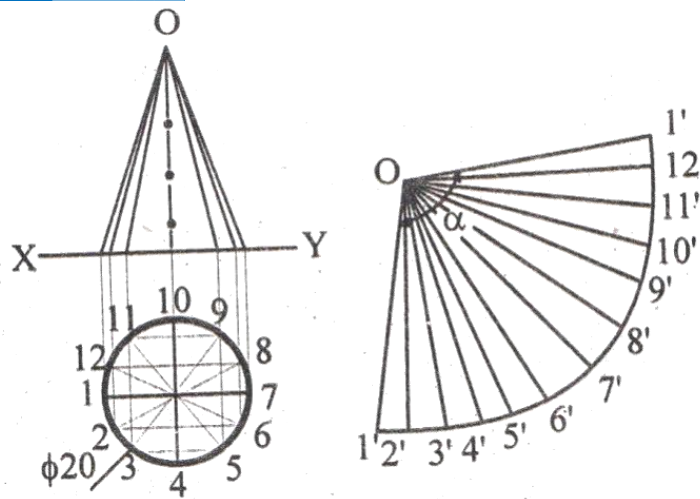


Figure 1.5

Construction:

1. Draw the XY line.
2. Draw the projections of the cone of diameter 20mm and axis height 30mm
3. Draw the circle into 12 equal parts named as 1, 2, 3, . . . etc.
4. For transferring slant edges to the development draw first parallel lines to the base upto true length line (O1').
5. Darken the developed portion of cone.

A right circular cone of diameter 60mm and height 70mm is cut by an inclined plane passing through the axis at a point 15mm above the base and inclined at 45 degree to HP and perpendicular to VP. Draw the development of lateral surface of the cone.

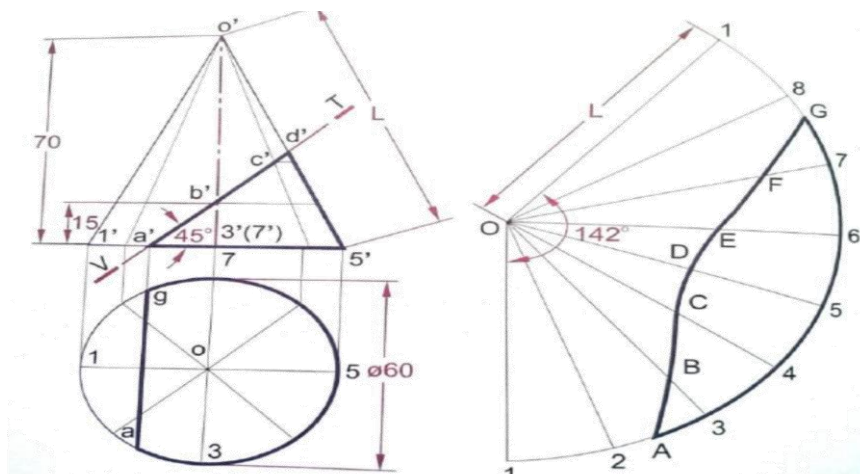


Figure 1.6

8. Assignments

S.No	Question	BL	CO
1	A cube of 35mm long edges is resting on HP on one of its faces with a vertical face inclined at 30° to VP. It is cut by a section plane parallel to VP and 9mm away from the axis and further away from VP. Draw its sectional front view and top view	1	4
2	i A triangular pyramid having base 40mm side and axis 50mm long is lying on HP on one of its faces with the axis parallel to VP. A section plane parallel to VP cuts the pyramid at a distance of 6mm from the axis. Draw its front view and sectional top view.	1	4
3	A square pyramid base 40mm side and axis 65mm long is lying on base on HP and all the edges of the base are equally inclined to VP. It is cut by a section plane perpendicular to VP and inclined at 45° to HP and bisecting the axis. Draw its sectional top view and true shape of the section and sectional side view.	1	4
4	A cylinder of 40mm in diameter and 60mm height and having its axis vertical is cut by a section plane perpendicular to VP, inclined at 45° to HP and intersecting the axis 32mm above the base. Draw its front view, sectional top view and true shape of the section.	1	4
5	A cone is resting on HP with its base of diameter 50mm and axis 70mm and it is cut by a section plane parallel to the base at a height of 20mm from the base. Draw its sectional top view.	1	4

Part B- Questions

S.No	Question	BL	CO
1	A cube of 35mm long edges is resting on HP on one of its faces with a vertical face inclined at 30° to VP. It is cut by a section plane parallel to VP and 9mm away from the axis and further away from VP. Draw its sectional front view and top view	1	4
2	A triangular pyramid having base 40mm side and axis 50mm long is lying on HP on one of its faces with the axis parallel to VP. A section plane parallel to VP cuts the pyramid at a distance of 6mm from the axis. Draw its front view and sectional top view.	1	4
3	A square pyramid base 40mm side and axis 65mm long is lying on base on HP and all the edges of the base are equally inclined to VP. It is cut by a section plane perpendicular to VP and inclined at 45° to HP and bisecting the axis. Draw its sectional top view and true shape of the section and sectional side view.	1	4
4	A cylinder of 40mm in diameter and 60mm height and having its axis vertical is cut by a section plane perpendicular to VP, inclined at 45° to HP and intersecting the axis 32mm above the base. Draw its front view, sectional top view and true shape of the section.	1	4
5	A cone is resting on HP with its base of diameter 50mm and axis 70mm and it is cut by a section plane parallel to the base at a height of 20mm from the base. Draw its sectional top view.	1	4

9. Supportive Online Certification Courses

1. Engineering drawing and computer graphics By Prof. Rajaram Lakkaraju, conducted by IIT Kharagpur – 12 weeks

10. Prescribed Textbooks & Reference Books**Textbook**

1. K.L. Narayana & P. Kanniah, Engineering Drawing, 3/e, SciTech Publishers, Chennai, 2012.
2. N.D. Bhatt, Engineering Drawing, 53/e, Charotar Publishers, 2016.

References:

1. Venugopal, Engineering Drawing and Graphics, 3/e, New Age Publishers, 2000.
2. K.C. John, Engineering Graphics, 2/e, PHI, 2013.

